

Ravada Koteswar Rao

Mechanical Design Engineer | Reverse Engineering | Product Development | Creo | Siemens NX

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PROFILE

Mechanical Design Engineer with 2+ years of experience in industrial equipment design, reverse engineering, and engineering support for Tata Steel projects. Proficient in Siemens NX, PTC Creo, AutoCAD, GD&T, Design for Manufacturability (DFM), and Design for Assembly (DFA). Experienced in developing 3D component and assembly models, manufacturing drawings, fabrication drawings, Bills of Materials (BOM), and engineering documentation. Strong understanding of machine design fundamentals, manufacturing processes, and design validation through site engineering and client collaboration.

EDUCATION

Bachelor of Technology

2021 – 2024

Anil Neerukonda Institute of Technology and Sciences
(Mechanical Engineering) CGPA : 8.48/10

Visakhapatnam, Andhra Pradesh

PROFESSIONAL EXPERIENCE

Praxsol Engineering Private Limited

03/2024 – Present

Junior Mechanical Design Engineer
Key Clients : TATA Steel, MVS ACMEI

Visakhapatnam, Andhra Pradesh

Design Engineering Activities

- Developed 200+ production-ready manufacturing, fabrication, and General Arrangement (GA) drawings for industrial mechanical components and assemblies, supporting production, installation, and maintenance.
- Developed accurate 3D component and assembly models using PTC Creo Parametric and Siemens NX.
- Generated and maintained Bills of Materials (BOM), ensuring proper documentation and traceability.
- Applied GD&T, Design for Manufacturability (DFM), and Design for Assembly (DFA) principles to improve manufacturability and assembly feasibility.
- Performed reverse engineering by transforming field measurements, engineering sketches, and existing components into accurate CAD models and manufacturing drawings.

Site Engineering Activities

- Worked on-site at Tata Steel Jamshedpur, supporting design and engineering activities.
- Executed reverse engineering by capturing design intent and critical dimensional data to develop production-ready CAD models and manufacturing drawings.
- Collaborated with clients and plant engineers in technical discussions to deliver engineering solutions.
- Led a team of four Graduate Engineer Trainees, ensuring quality and timely delivery of design outputs.
- Ensured accurate alignment of drawings with actual plant conditions through on-site verification, measurements, and installation feasibility checks.

SKILLS

CAD & PLM Tools

- Siemens NX
- PTC Creo Parametric
- AutoCAD
- CATIA V5
- Teamcenter (Self-Learning)

Standards & Documentation

- GD&T Fundamentals
- Indian Standards (IS)
- Bill of Materials (BOM) Creation & Documentation
- Design for Manufacturability (DFM)
- Design for Assembly (DFA)

Design & Engineering

- Reverse Engineering
- 3D CAD Modeling
- Manufacturing & Fabrication Drawings
- General Arrangement (GA) Drawings

PROJECTS

Reverse Engineering & Product Development

04/2024 – Present

Tools : PTC Creo, Siemens NX, AutoCAD Mechanical

- Executed end-to-end reverse engineering, transforming field measurements, engineering sketches, and existing components into accurate CAD models and manufacturing drawings.
- Developed 3D CAD models and manufacturing drawings for industrial, automobile, and locomotive components, including shafts, gears, gearboxes, pulleys, sprockets, plunger blocks, base frames, casting, sheet metal, and fabricated components.
- Prepared manufacturing, fabrication, and General Arrangement (GA) drawings, ensuring dimensional accuracy, fitment, manufacturability, and installation feasibility.
- Collaborated with clients, plant engineers, and manufacturing teams during technical discussions and engineering reviews while applying GD&T and Design for Manufacturability (DFM) principles.

Web Centric Design for Smart Gate Systems

12/2023 – 03/2024

- Developed a web-based system for smart gate design, enabling efficient design iterations and usability with integrated mobile control and monitoring.

CERTIFICATIONS

Siemens NX

Linkedin Learning

CATIA V5 (CAD)

Infosys Springboard.

Introduction to Geometric Dimensioning and Tolerancing (GD&T)

Linkedin Learning

Additive Manufacturing Manufacturability : Laser Powder Bed Fusion(L-PBF)

American Society of Mechanical Engineers (ASME).

LANGUAGES

- English
- Hindi
- Telugu

DECLARATION

I hereby confirm that the information given is true to my knowledge and belief.